Land Usage Classification in Agriculture

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1 Abstract

In this laboratory, we explore how land use types can be classified based on aerial and satellite photographs of the earth's surface. We learn to build our own photo masks based on ready-made masks and display them on the screen. The laboratory focuses on transforming images into datasets, building classification models, and creating land use masks based on classifier predictions.

2 Introduction

Determining how land is used is a significant challenge today, as improper and illegal use can lead to both economic and natural disasters. One approach to assess land use involves analyzing aerial and satellite images of the earth's surface. A major challenge lies in constructing mathematical models that can determine land use types based on colors in these images. With ready-made photos and masks of land use, artificial intelligence and big data methods can be employed to build classification models.

In this lab, we learn how to upload photos, transform pixels and colors into a structured dataset. We then build a classifier and predict land use masks based on it. The statistical data used in this laboratory is obtained from Kaggle under the CC0: Public Domain license.

2.1 Objectives

After completing this lab, you will be able to:

- Download and transform images into analyzable formats.
- Create a dataset from the colors present in images.
- Build a classification model to predict land use types.
- Construct land use masks based on classifier predictions.

2.2 Prerequisites

This lab requires familiarity with the following Python libraries at specified proficiency levels:

- Python - basic level
- numpy - middle level
- Seaborn - basic level
- Matplotlib - basic level
- scikit-learn - middle level
- pandas - basic level

3 Materials and Methods

The laboratory consists of four main stages:

1. Download and visualization of images
2. Dataset creation
3. Classification model creation
4. Create your own mask of land use

The dataset used in this lab comes from semantic segmentation of aerial imagery and contains 72 images with corresponding masks, though a subset of 9 images is used for practical training purposes due to computational constraints.

3.1 Download and Visualization of Images

A function is created to load and display images and their corresponding masks from specified directories. The function performs the following operations:

1. Loads a JSON file containing class descriptions and displays the classes
2. Downloads all images from a specific directory using Image.open()
3. Displays the last image along with its corresponding mask
4. Forms and returns a dataset as an array of tuples [image, mask]

The dataset includes 6 land use classes, each represented by a distinct color in the mask files. Due to the large size that a full pixel-level dataset would create, only 9 images are selected for processing.

3.2 Dataset Creation

Each image consists of multiple points, where each point is represented by an RGB tuple (red, green, blue). Colors are represented as integers in the range [0, 255]. An image is structured as a 3D array (height, width, color channels).

To establish the relationship between color and land use class, each image is transformed into a dataset of the form (r, g, b) $\rightarrow$ class. This transformation involves:

- Converting images to color matrices using np.asarray()
- Flattening these matrices using np.flatten()
- Converting mask colors from RGB to hexadecimal format using custom rgb2hex() functions

The training dataset consists of 4 images, while the test dataset contains 5 images. All columns in the initial dataset have object type, requiring conversion of the HEX color column to categorical data type for proper analysis.

The dataset is visualized using Seaborn countplots to examine the distribution of pixel classes across the 6 land use categories. Both training and test datasets show similar class distributions, making them suitable for model training and evaluation.

3.3 Classification Model Creation

For mask analysis, the sklearn.linear_model.LogisticRegression() classifier is employed. This classifier offers fast and simple implementation, suitable for initial experimentation with pixel-based classification.

The model uses the first three columns (RGB values of image pixels) as input features, and the HEX color column from mask images as the target output. Training and evaluation utilize the fit() and score() functions. The ConfusionMatrixDisplay is used for detailed performance analysis.

To handle potential class imbalance, class weighting is applied, and a pipeline is constructed with preprocessing steps including StandardScaler. The target variable is encoded using LabelEncoder to convert categorical class labels into numerical format.

The model demonstrates reasonable accuracy with minimal difference between training and test performance, indicating good generalization. Further accuracy improvements would require expanding the dataset with more images.

3.4 Create Your Own Mask of Land Use

The final stage involves building custom land use masks based on the trained classifier model. This process includes:

- Selecting test images from the downloaded data list
- Building a dataset for these test images
- Using the trained model to predict hex colors for each pixel
- Reshaping the 1D prediction array into a 2D matrix matching original image dimensions
- Converting hex color predictions to RGB values for visualization

The conversion from class predictions to RGB images requires careful handling of colormaps. A dictionary mapping class indices to hex colors is created, and functions are implemented to convert hex codes to RGB tuples suitable for display.

The generated mask is compared with the original mask to assess model performance. Both vectorized and loop-based approaches are demonstrated for creating RGB images from class predictions, with vectorized methods offering significant performance advantages for larger images.

The comparison typically shows strong similarity between predicted and actual masks, with discrepancies attributable to the limited training dataset size. This demonstrates that increasing dataset size would likely improve accuracy further.

4 Results

4.1 Dataset Characteristics

The dataset loading process reveals 6 distinct land use classes, each with corresponding color codes in mask files. The subset of 9 images is successfully loaded and split into training (4 images) and test (5 images) sets.

The training dataset, after transformation, contains 2,050,681 rows representing individual pixels, while the test dataset contains 2,566,340 rows. All columns initially have object type, with the HEX column subsequently converted to categorical type for analysis.

4.2 Class Distribution

Visualization of class distributions using count plots confirms that both training and test datasets contain similar proportions of the 6 land use classes. This balanced distribution supports effective model training and evaluation.

4.3 Model Performance

The logistic regression classifier achieves consistent accuracy on both training and test datasets, with minimal performance gap indicating good generalization. Confusion matrix analysis provides detailed insight into class-wise prediction performance, revealing which land use categories are most reliably identified and where confusion occurs between similar classes.

4.4 Mask Generation

Custom masks generated from test images closely resemble the original masks, successfully identifying major land use regions. The vectorized approach for converting class predictions to RGB images proves efficient and accurate.

The minimal faithful subgraph concept from GNN analysis finds analogy here in the minimal set of pixels needed to preserve classification accuracy, demonstrating that not all pixels are equally important for determining land use type.

5 Conclusion

In this laboratory, we have studied how to create an expert system based on classifiers for land use classification from aerial and satellite imagery. The key accomplishments include:

- Successfully downloading and transforming images into analyzable datasets
- Mastering the extraction of image colors and construction of training and testing sets
- Building and evaluating a logistic regression classifier for pixel-level land use prediction
- Creating custom land use masks based on classifier predictions
- Calculating and interpreting accuracy metrics

The principles demonstrated in this lab extend to any type of image classification task and any categories of land use. The strong similarity between predicted and actual masks confirms the viability of this approach, while the observed performance gap indicates that expanding the training dataset would further improve accuracy.

This laboratory provides foundational skills for applying machine learning to remote sensing and agricultural monitoring applications, where automated land use classification supports informed decision-making and resource management.

6 References

1. Kaggle Dataset: Semantic Segmentation of Aerial Imagery. Available at: `https://www.kaggle.com/humansintheloop/semantic-segmentation-of-aerial-imagery`. Licensed under CC0: Public Domain.
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End of Report

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